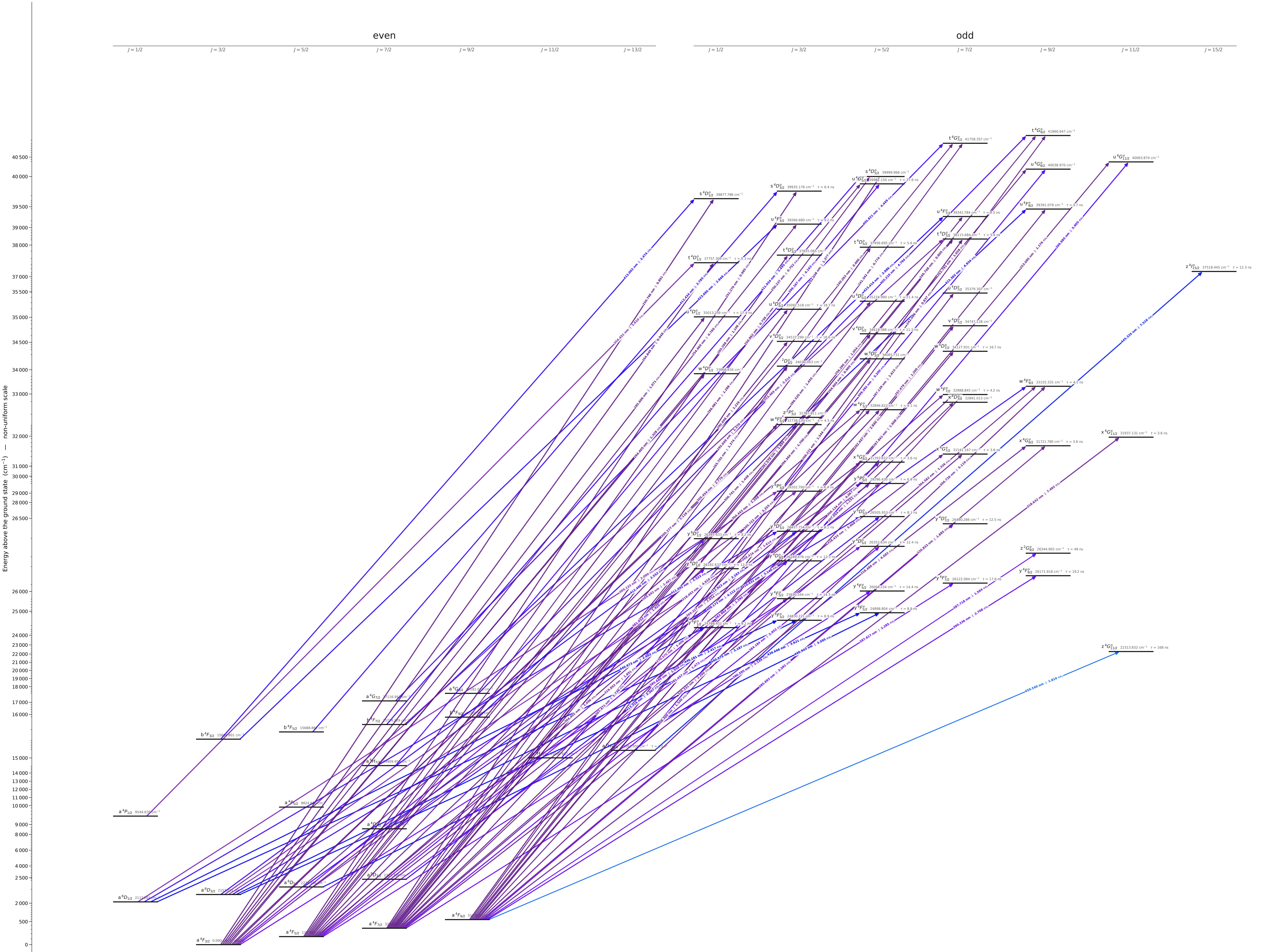




Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er J'] (e.a.)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁴ H _{1/2} - z ⁴ F _{5/2}	445.3255	445.2005	673.1985	7.516	8.100e+07	12.94	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁵¹V hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
y ⁶ F _{5/2} 99(30)	-	-	F=1: -1112.98 F=2: -915.12 F=3: -618.32 F=4: -222.60 F=5: +272.06 F=6: +865.65
y ⁶ D _{5/2} 15(30)	-	-	F=1: -168.63 F=2: -138.65 F=3: -93.69 F=4: -33.73 F=5: +41.22 F=6: +131.16
y ⁶ P _{3/2} 540(60)	-	-	F=2: -3642.48 F=3: -2023.60 F=4: +134.91 F=5: +2833.04
y ⁶ P _{5/2} 390(60)	-	-	F=1: -4384.46 F=2: -3605.00 F=3: -2435.81 F=4: -876.89 F=5: +1071.76 F=6: +3410.14
x ⁴ G _{5/2} 603(30)	-	-	F=1: -6779.06 F=2: -5573.89 F=3: -3766.14 F=4: -1355.81 F=5: +1657.10 F=6: +5272.60
x ⁴ G _{7/2} 293(30)	-	-	F=0: -4580.08 F=1: -4289.28 F=2: -3707.69 F=3: -2833.29 F=4: -1672.09 F=5: -218.10 F=6: +1526.69 F=7: +3562.29
x ⁴ G _{9/2} 159(30)	-	-	F=1: -3058.63 F=2: -2740.85 F=3: -2264.18 F=4: -1629.62 F=5: -834.17 F=6: +119.17 F=7: +1231.40 F=8: +2502.52
x ⁴ G _{11/2} 81(30)	-	-	F=2: -1841.48 F=3: -1598.64 F=4: -1274.87 F=5: -870.15 F=6: -384.48 F=7: +182.12 F=8: +829.68 F=9: +1558.17
w ⁴ F _{5/2} 258(30)	-	-	F=1: -2900.50 F=2: -2384.85 F=3: -1611.39 F=4: -580.10 F=5: +709.01 F=6: +2255.94
w ⁴ F _{7/2} 108(30)	-	-	F=0: -1699.82 F=1: -1591.89 F=2: -1376.04 F=3: -1052.27 F=4: -620.57 F=5: -80.94 F=6: +566.61 F=7: +1322.08
z ⁴ F _{5/2} 183(30)	-	-	F=4: -5440.47 F=5: -4526.11 F=6: -3428.87 F=7: -2148.76 F=8: -685.77 F=9: +960.08 F=10: +2788.81 F=11: +4800.42
t ⁴ D _{5/2} 213(30)	-	-	F=2: -1436.76 F=3: -798.20 F=4: +53.21 F=5: +1117.48
t ⁴ D _{7/2} 144(30)	-	-	F=0: -2266.43 F=1: -2122.53 F=2: -1834.72 F=3: -1403.03 F=4: -827.42 F=5: -107.92 F=6: +755.48 F=7: +1762.78
u ⁴ F _{8/2} 285(30)	-	-	F=1: -5482.46 F=2: -4912.85 F=3: -4058.44 F=4: -2919.23 F=5: -1495.22 F=6: +213.60 F=7: +2207.22 F=8: +4485.65

Measured A, B constants; sources in the data file.